Cuando OXXO llega a la cuadra:

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https://github.com/sophiearisti/proyecto_econometria_avanzada_real.git

1 Abstract

2 Acknowledgements

3 Introduction

The rapid expansion of modern retail chains in Latin America after the 2019 COVID-19 pandemic has transformed urban commercial dynamics and people's daily routines in the main cities of the region (Antom Knowledge, 2024; Company, 2024; Sánchez-Díaz et al., 2021). These establishments strategically locates in high-traffic areas, residential neighborhoods, and commercial corridors. Additionally, their 24-hour establishments offer basic goods, financial services, and bill payment facilities, attracting diverse customer flows throughout the day and night. Given their expanding footprint within the urban landscape, these retail hubs have naturally emerged as magnets of economic, urban and pedestrian activity (Dugato et al., 2026; Marcos, 2022).

Convenience stores' growing commercial importance has prompted economic research to primarily focus on the impact of these establishments on retail businesses, consumer behavior, and the labor market (Delgado et al., 2024). However, beyond their economic role, their physical and operational characteristics place them in a contested position within the literature on urban crime.

Depending on the context, these stores are seen as *crime generators* by increasing foot traffic and creating incidental criminal opportunities (Dugato et al., 2026). Also, they have been acknowledged as robbery prone commercial establishments as they have available cash on hand (Wellford et al., 1997). Conversely, they are perceived as crime combators by providing natural surveillance and increasing 'eyes on the street' through extended operating hours and improved lighting (Davis et al., 2021). This duality highlights the necessity of analysing their role on their surrounding criminal environment, as targeting the aspects that facilitate criminal behaviour is often more effective for long-term crime reduction than traditional repression (Dugato et al., 2026).

OXXO, a Mexican retail chain now owned by FEMSA, represents a valuable case study for this trend, having intensively proliferated across Mexico, Colombia, Peru, and Brazil with more than 25,000 locations. In Colombia specifically, the chain operates more than 500 stores with over 300 located in the capital, Bogotá Bogotá (Godoy, 2025). Since more than 50% of these establishments are located in Bogotá, the city serves as a unique laboratory for understanding the spatial relationship between criminal activity and OXXO's presence.

Additionally, researching this phenomenon is crucial for the city. Although Bogotá has experienced significant crime reduction since 1990s (Sánchez et al., 2003), it is still a major public policy concern as high volumes of personal theft and a rising public perception of insecurity continue to affect the daily lives of its citizens (Cámara de Comercio de Bogotá, 2026). Additionally, crime remains spatially concentrated, with substantial variation across zones driven by socioeconomic stratification, land use, and policing intensity (Enriquez et al., 2014; Giménez-Santana et al., 2018), wich underscores the importance of evaluationg how the introduction of these modern retail chains could reconfigure Bogotá's urban criminal environment.

Thus, in this paper, I aim to explore the causal effect of OXXO openings on local crime activity in

Bogotá and analyze if effects vary by crime type and time since opening. To estimate this impact, I use the public georeferenced crime data stored in the Statistical, Criminal, Contraventional, and Operational Information System of the National Police (SIEDCO). I create a quarterly panel aggregated at the level of the Zonal Planning Unit (UPZ), which serves as a sub-district administrative division in Bogotá. I also employ Two-Way Fixed Effects (TWFE) and the Callaway and Sant'Anna (C&S) estimator, along with their corresponding event study methods, to capture the dynamic evolution of the treatment effect over time.

Despite extensive research on urban crime in Bogotá, no study has systematically examined the causal impact of convenience store proliferation, particularly OXXO, on local crime patterns. Existing research has focused on public transport infrastructure (Bacares, 2013; Bacares et al., 2013a; García et al., 2005), large-scale policing interventions (Blattman et al., 2017, 2021), alcohol sales restrictions (Mello et al., 2013), and socioeconomic determinants (Escobar, 2012). One recent study examined low-cost supermarkets or hard-discount stores (Tiendas D1) and crime (López et al., 2024), but OXXO represents a distinct retail model characterized by 24-hour operations and a higher density of "grab-and-go" items, which may create different criminal opportunities compared to the limited operating hours of hard-discount formats.

The remainder of this paper is organized as follows:

4 Background

4.1 The expansion of OXXO in Bogotá

OXXO is one of the most prominent convenience store brands in Latin America, and since its arrival to Colombia in 2009 it has consolidated itself as one of the main convenience store chains in the country. Although OXXO has been operating in Colombia for almost 17 years, it only gained momentum after the pandemic. In fact, in Bogota the number of establishments skyrocketed from 82 in 2020 to 384 in 2025. This represents an increment of over 368% in only five years. Figure 1 depicts its accelerated growth in Bogotá, as it shows the number of OXXO locations in the capital by year since its arrival.

OXXO's main objective is to be part of people's lifestyles by offering an ample variety of products and services that can be useful and practical for their daily lives (OXXO Colombia, 2026b). Another distinctive characteristic is that 95% of its establishments function in a 24-hour format (OXXO Colombia, 2026a), providing a competitive advantage by being the primary option of consumers whenever they need to satisfy spontaneous consumption needs, regardless of the hour. These well-defined competitive objectives and inherent brand characteristics have allowed OXXO to proliferate rapidly throughout Bogotá, where it has been widely embraced by the local population.

In line with this strategy, this modern retail chain began its expansion in the northern part of the city, later expanding throughout the eastern region. This strategic behavior responds to the inherent characteristics of the northeastern neighborhoods of Bogotá, where a higher concentration of formal

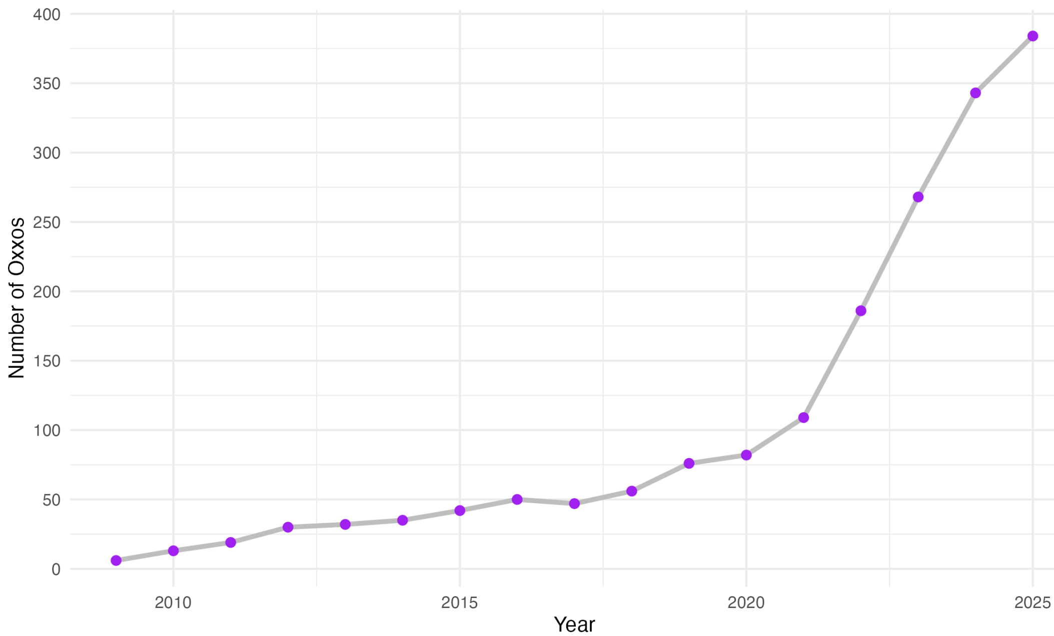


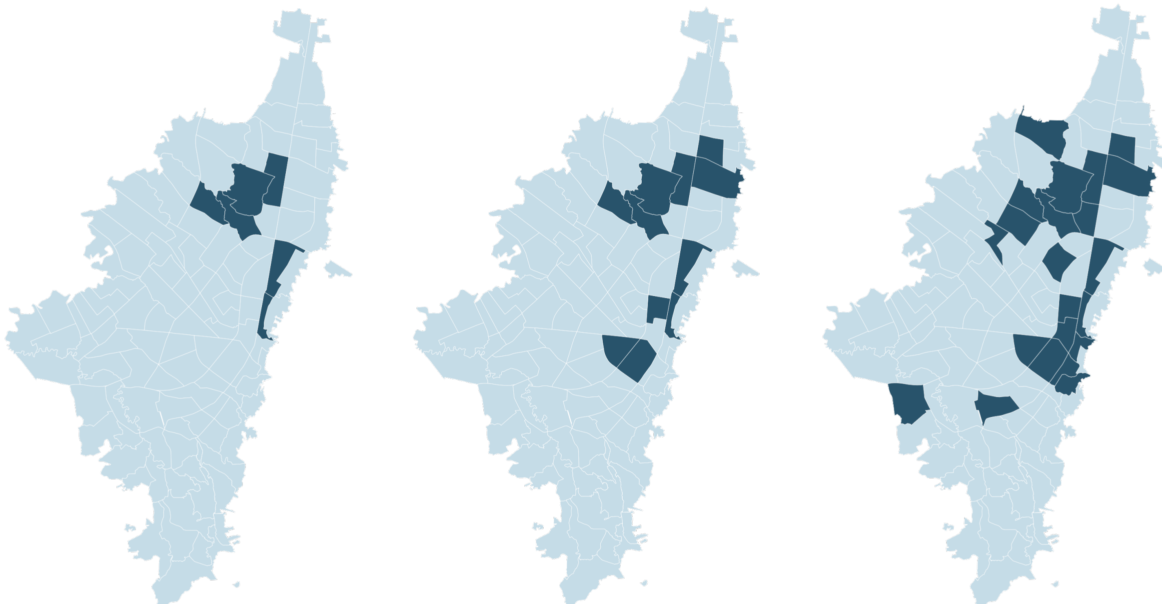
Figure 1: **Yearly evolution of the number of OXXOs in Bogotá**
Note: BLAH BLAH

employment and high-income residential clusters provide an ideal environment for the convenience store model (citation).

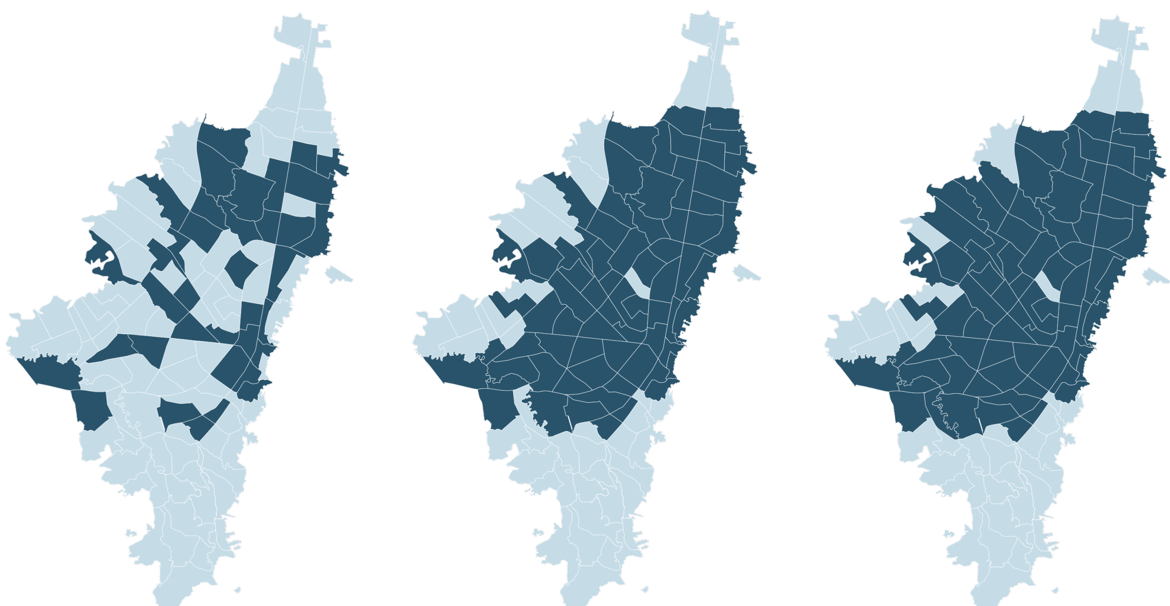
Figure 2 presents this evolution, and shows that now, OXXO is present in almost 70% of all the city´s Zonal Planning Units (UPZ). UPZs are analysis units that group neighborhoods based on their socioeconomic stratification, commercial, land use, and housing characteristics, and living conditions of their inhabitants (Alcaldía Mayor de Bogotá, 2021). Currently, the only areas without OXXO establishments are in the far southern reaches of Bogotá, which correspods to UPZs with low levels of economic performance, socioeconomic stratification, urban transport connection, and are blocks that continue to relly heavily in traditional neighborhood stores (citation).

As noted, OXXO's stores are not randomly located. They are centrally administered by FEMSA (meaning the brand does not operate under a franchise model) with the purpose of standardizing processes, logistics, services, and products (OXXO Colombia, 2026a). This means that each location is carefully planned to respond to the overall interests of the company. In fact, if we attentively analyze how these establishments are clustered around Bogotá, we can identify that they tend to localize in areas where there is higher traffic and spatial accessibility. This helps OXXO to maximize its market reach and ensure high levels of consumer visibility.

Figure 3 graphically demonstrates this correlation by plotting the city's arterial roads, main avenues, and Transmilenio stations (Bogotá's Bus Rapid Transit system); it shows that by 2025, OXXO locations tend to group along these major transport corridors.



(a) 2009 (b) 2011 (c) 2015



(d) 2019 (e) 2023 (f) 2025

Figure 2: **Evolution of the presence of OXXO by UPZ**

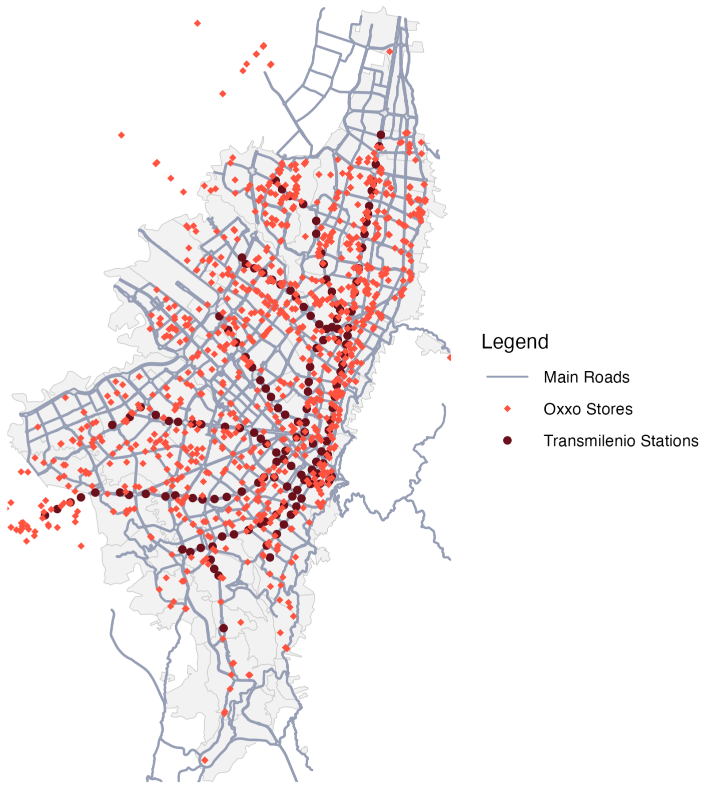


Figure 3: **Bogota's OXXO stores, Transmilenio stations and main roads in 2025**
Note: BLAH BLAH

With these in mind, it is reasonable to think there may be a correlation between urban crime and modern convenience stores chains, especially in the case of OXXO. Locating in areas with high foot traffic means more potential targets for crimes such as robbery or assault. Also, due to their 24-hour format and alcohol sales during the night, OXXO locations serve as informal gathering points that may escalate into conflicts or provide opportunities for illicit activities. Additionally, locations near major transit areas and public transport stops increase accessibility for both legitimate customers and offenders (Dugato et al., 2026; Feng et al., 2019; Shin, 2024).

4.2 Criminality in Bogotá

Between the 80s and 90s Bogota was recognized as one of the most insecure cities of Colombia, in fact, in 1991 it reached an annual crime rate of 1,287 crimes for each 100 thousand inhabitants (Rodríguez González, 2017). During this time the capital lived an upswing of narcoterrorism and the assassination of political leaders. However, at the end of the 90s and beginning of the 2000s, crime started to decrease; for example, between 1993 and 2002 Bogotá saw an estimated 65% decrease in the homicide rate, while robbery fell by about 33%. This shift was driven by citizen culture programs, large-scale urban renewal projects, and robust institutional reforms implemented by the majors of this period (Mockus, 2012; Sánchez et al., 2003). Finally, in past years urban crime has shown resistance to keep declining. After the 2019 covid pandemic criminality had a significant rebound, with personal theft surpassing pre-pandemic levels. By 2025, despite institutional efforts, the city's crime rates have shown resistance to reduce, particularly regarding homicides and robberies hl(citation).

Currently, Bogota's crime, as any other city, clusters in specific zones. The most insecure Localities (a unit of territorial division that gathers many UPZ) are located in the southeastern region of Bogota, wich correlates with a high presence of criminal organizations, informality, and deficencies of human development (ProBogotá Región, 2024). However, this concentration does not imply that all crime categories peak in the same areas. While homicide rates tend to follow this general rule in the city, personal theft exhibits a distinct geographic distribution, primarily clustering in the city's northeastern sector (ProBogotá Región, 2023).

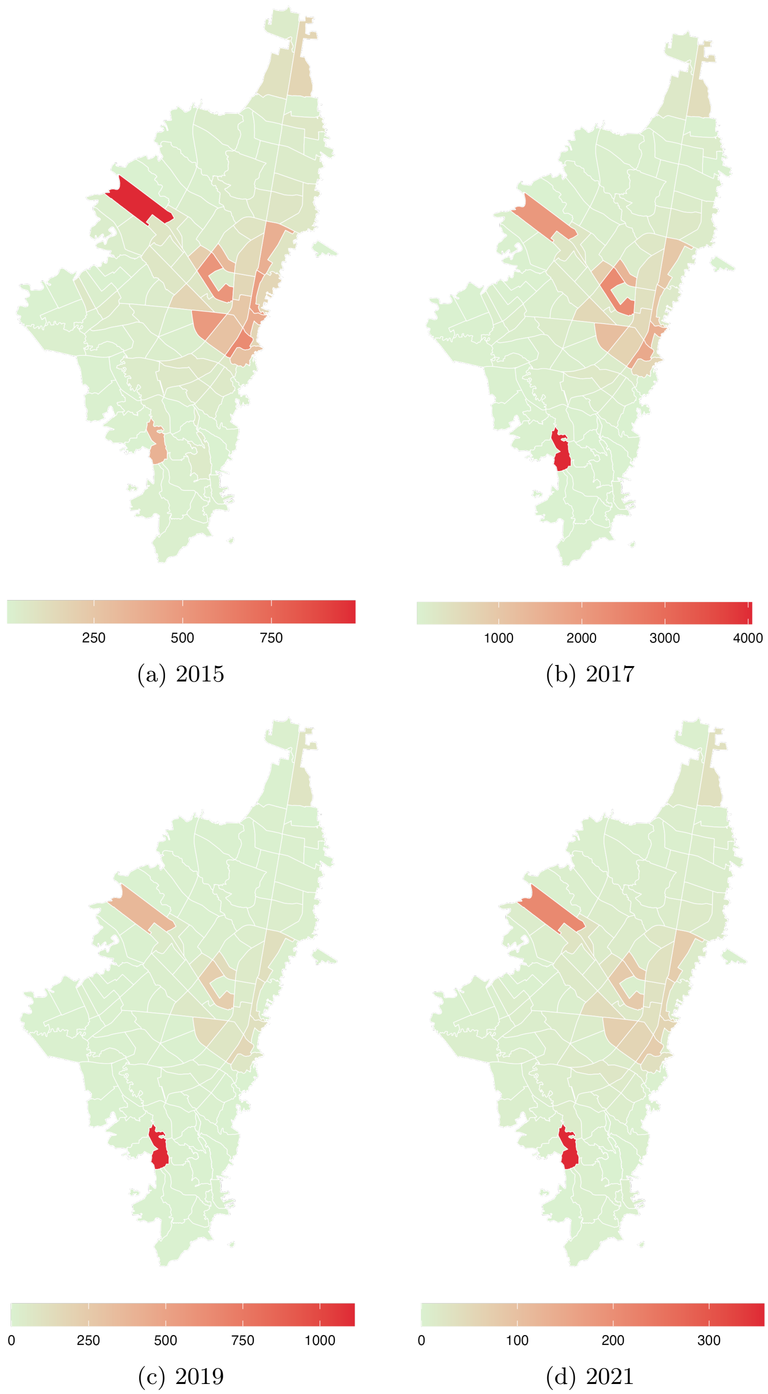
Figure 3 shows this behaviour as it displays how personal theft rates distribute throughout Bogotá´s UPZs over the years. These maps also highlight that between 2015 and 2021 (and even later, as it can be noted in ProBogota's annual security informs hl(PIE DE PAGINA)) this tendency has not drastically changed, meaning that there are environmental characteristics that keep favoring robberies or attracting offenders. Other characteristics that these zones have are high pedestrian throughput, the density of commercial land use, and the proximity to the city's main transit arteries.

Factors that favor crime clustering must be taken into account because if OXXO stores also locate in inherent high crime areas, it will be important to evaluate whether OXXO efectively causes an increment in crime or if both OXXO and criminals are simply attracted to the same high-activity "nodes."

5 Related Literature

In the context of urban economics, spatial econometrics and crime literature, various studies have explored how built environment features and infrastructure shape crime activity in cities. A number of empirical analyses have investigated this by exploring the causal impact of urban public transit on crime. Xu et al., 2021, for example, assessed the impact of two new subway lines implemented in 2017 in ZG city, China, on thefts in the surrounding areas. By employing a Difference-in-Differences (DiD) framework, they found that in the short run, theft increased by 15%. Others, on the other hand, have ventured to explore the paper of new urban policies and infrastructure interventions on this topic. Domínguez and Asahi, 2022 paper is one case, as they study the causal effect of ambient light on crime in Santiago, Chile, by using the "Daylight Saving Time" policy as a natural experiment. Due to this policy, they could find that an increase of one hour of ambient light exposure between 7 and 9 reduced thefts by 30%. These studies empahsize on XXXXX (give emphasis on a crime literature concept)

Retail chains can be considered as built environment features that may influence urban crime activity. While different studies have extensively explored their role, these previous inquiries have their own limitations. A substantial body of research dates back to the pre-2000s era, meaning that their



(a) 2015 (b) 2017

(c) 2019 (d) 2021

Figure 4: **Evolution of theft to people rates by UPZ**

findings do not contemplate the current urban dynamics of modern cities. Nonetheless, these foundational studies provide a critical perspective on the underlying mechanics of crime. For instance, early investigations focused on identifying the characteristics that influenced the propensity of convenience stores to be targeted for robbery. They identified that stores that operated for longer hours at night were more likely to be robbed, as there are fewer people around (D'Alessio & Stolzenberg, 1990). Others found that these types of stores were more prone to be robbed due to the lack of visible security such as police presence, security guards, or effective alarm systems (Dugato et al., 2026; Wellford et al., 1997). Although useful, they fail to account for convenience stores' role on crime of their immediate surroundings.

On the other hand, other recent publications have examined the influence of analogous commercial establishments on urban crime. While some of these studies do not focus exclusively on convenience stores, they utilize other retail outlets with similar operational profiles which can serve as proxies for what this thesis aims to investigate. Askey et al. (2018) is and example as the authors pooled data from fast-food restaurants and convenience stores in Seattle. They used sales revenue as a proxy for human activity to count the different levels of operational intensity across locations, and aimed to explain crime patterns in the immediate surroundings of these establishments. Their results indicate that stores with higher average sales were associated with with a 7.4% increase in expected crime counts. Conversely, Feng et al., 2019 contribute to this discussion by finding that alcohol outlets influenced the likelihood of aggravated assault and street robbery in New York City. Their findings reveal differential impacts depending on the type of outlet (if it is a bar, a grocery store, or other similar establishment); for instance, the authors highlighted that in Manhattan and the Bronx, proximity within two blocks of a grocery store increased the relative risk of street robbery by more than sixfold.

Another relevant investigation is the one of (Shin, 2024). Shin used a DiD approach to invetigate the impact of dollar stores on localized violent crime patterns in Chicago. To ensure the treatment and control groups were as homogeneous as possible, the author created localized areas using a 250-feet buffer for the treatment group and a 250-to-354-feet outer ring as the control. At the end, this research found that store openings lead to a 21.% increase in violent crime (0.17 incidents per quarter), a result primarily driven by a 38% surge in robberies.

Although these studies offer a useful baseline of how convenience stores, and specifically OXXO, impact nearby crime, they present an incomplete perspective as they exclusively explored this nexus in cities of high-income nations. To build a more robust theory, these hypotheses must be tested within Latin American cities which have particular urban dynamics. Many papers argue that convenience stores are normally located in socio-economically disadvantaged areas where poverty, social disorganization, and lack of community resources exacerbate crime risks (Dugato et al., 2026; Shin, 2024), but in cities like Bogotá the situation is not analogous. OXXO in Bogotá, for instance, does not follow this rule as it instead prioritizes to open locations in middle to high income neighborhoods, and places with high commercial activity and influx of people.

In Latin America, the economic literature regarding convenience stores remains limited. Recent investigations have explored their causal role on various labor market indicators (Delgado et al., 2024; Marcos, 2022). However, despite this growing interest in labor dynamics, the spatial relationship between these establishments and urban security in the region has been largely overlooked. Specifically in Bogotá crime literature has primarily focused on public transport, general socioeconomic urban caracteristics, and policy intervention impact on crime (Bacares et al., 2013b; Moreno García, 2005).

The investigations that have actually explored this topic are counted. For example, Jaramillo López et al., 2021 used the same method as (Shin, 2024) to study the effect of D1 hard discount stores on the crime around the streets where they open in Bogotá. Although D1s are not convenience stores per se, they attract a similar type of costumers and and often share similar spatial locations with OXXO outlets. By analyzing the period of 2016 to 2019 and employing the same information utilized in this thesis (SIEDCO), the authors found that D1 openings caused a 33.4% average reduction in crime density per block. They attributed this impact to the 'eyes on the street' phenomenon, suggesting that the increased pedestrian flow acts as a form of informal guardianship that deters criminal activity (Jacobs, 1961; Jaramillo López et al., 2021).

6 Data used and unit of analysis

For this thesis I use UPZs as the unit of analysis. UPZs work as a feasible division of Bogotá as the disctrict maintains continuity and consistency of data across the years, and security interventions are normally focalized by UPZ (cita). Also, these Units maintain urbanistic and socioeconomic homogeneity by grouping similar neighborhoods, and are the middle point between Localities and Street blocks (manzanas).

Meanwhile, the results of this study employ four main data sources (see appendix A). For measuring crime rate in Bogotá I used the georeferenced crime events stored in SIEDCO [1] spanning from 2018 to 2022. SIEDCO is Colombia's official Police database that consolidate real-time citizen reports and administrative records of criminal activity. Here, for each incident recorded the system saves details including type of crime (e.g., homicide, personal robbery, commercial burglary, or cell phone theft), the precise geographic coordinates of the event, date and time of the incident, and the victim's gender. With this information, I aggregated the crime data at the UPZ-year level, categorizing incidents by offense type. To ensure comparability across years and Units, I normalized these counts using annual population estimates for each UPZ. This allowed for the construction of a crime rate per 10,000 inhabitants, as shown in this equation:

$$crime\_type\_index_{upz,year} = \frac{total\_incidents_{upz,year}}{population_{upz,year}}; \quad \forall upz, year \tag{1}$$

The period 2018 to 2022 is adequate for the analysis as they englobe the years in which OXXO had

[1] Access at https://oaiee.scj.gov.co/agc/rest/services/Tematicos_Pub/SIEDCO_Delitos_Pub/FeatureServer

its explosive expansion across Bogota. This timeframe ensures sufficient temporal and spatial variability, meaing it allows for a higher proportion of UPZs to enter treatment during the study period compared to those always treated. Prior to 2018, the lack of new store openings in other UPZ would result in an insufficient number of newly treated units, limiting the identifying variation necessary for using TWFE.

Additionally, from the Registro Único Empresarial y Social (RUES) [2], the central business information system in Colombia that contains all the records of companies and merchants registered by the chambers of commerce in the country, I identified the opening and closing year of each OXXO. Additionally, with Google Maps API I obtained the precise georeferenced location of each store. Concerns may arise on how precise was this match of each store with its correct location, specially for the ones that no lorgers exist. I manually checked the validity of each match with official OXXO publications, and as a result XX% of establishments were properly matched.

I also did this same process with D1, Justo&Bueno and ARA hard discount chains. The reason why I also took into account these brands is because they often share similar spatial locations with OXXO outlets; in fact, retailers tend to cluster in certain areas to take advantage of agglomeration benefits (Konishi, 2005; Seong et al., 2022). Given that prior research suggests these establishments may influence both crime and commercial activity (Delgado et al., 2024; Jaramillo López et al., 2021), their inclusion as control variables is important to evaluate the validity of the results.

Finally, I used the Quality of Life Survey of Bogotá 2007 [3] to analyze the initial socioeconomic characteristcs of never treated and treated UPZ before OXXO arrived in 2009. And with Mapas-Bogotá I obtained the fixed variable socioeocnomic and environmental controls at the UPZ level just to explore how are OXXOs not randomly located, as these will be inhenrently controlled by TWFE.

[2] Access at https://www.rues.org.co

[3] Access at https://www.dane.gov.co/index.php/estadisticas-por-tema/salud/calidad-de-vida-ecv/encuesta-nacional-de-calidad-de-vida-2007-bogota

7 Descriptive Statistics

8 Control Variables

Table 1: Differences of means between treated and never treated for fixed controls

	Never treated	**Treated**	**Difference of means**
Baseline			
Inhabitants by locality in 2007	503.534	482.463	-21.071
	(38.331)	(43.823)	
Household size 2007	3.714	3.335	-0.380^{***}
	(0.049)	(0.048)	
Index of quality of life 2007	89.121	91.724	2.603^{***}
	(0.151)	(0.216)	
Average monthly expenditure 2007	$8.97e+05$	$1.38e+06$	$4.82e+05^{***}$
	(55836.359)	(57218.217)	
Fixed controls			
Average stratum	2.162	3.352	1.190^{***}
	(0.135)	(0.119)	
Nearby transmilenio stations	2.339	6.067	3.727^{***}
	(0.404)	(0.544)	
Access to arterial roads	29.286	44.200	14.914^{***}
	(4.490)	(2.738)	
UPZs	56	60	116

Note: Significance: *** p<0.01, ** p<0.05, * p<0.1.

9 Empirical strategy

10 Results

11 Discussion

12 Robustness

spill over effects, placebo tests with other crime types, etc. Vary the buffers fo the spill over effects, etc. propensity score matching treat controls and treated groups as buffers

13 Conclusions

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14 Appendix

A Detailed description of variables

Table A.1: Data sources

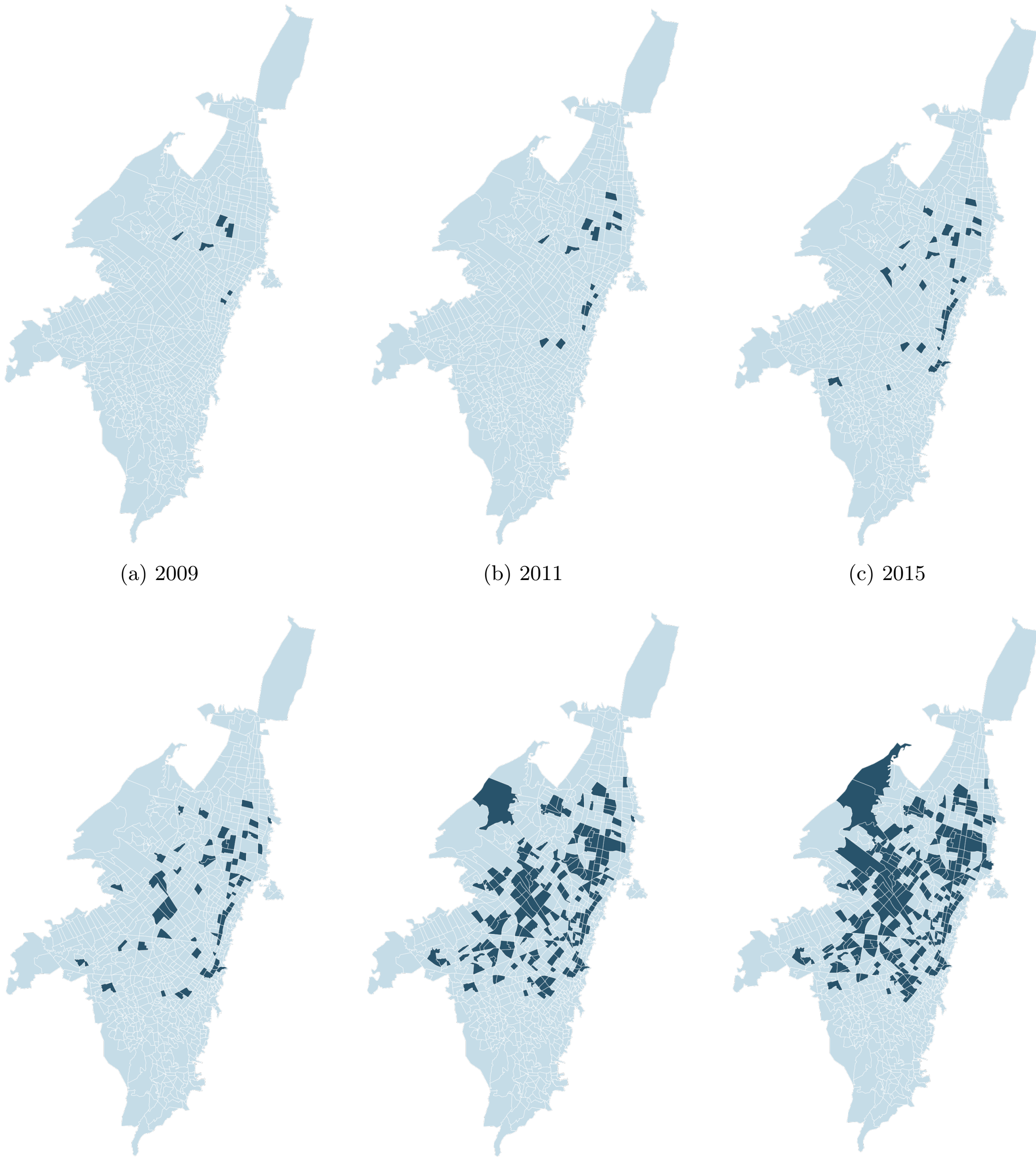
Variable	Description	Data source
Dependent Variable		
Proportion of independents	Proportion of independents in the ZAT that year	Movility surveys (2011, 2015, 2019, 2023)
Variable de tratamiento escalonado		
Tiene OXXO (= 1)	Si el ZAT tiene OXXO(s) ese año	RUES y Google Maps API
Controles resagados de tiendas de cadena		
Tiene D1s (=1)	El ZAT tiene D1(s) el periodo anterior al año analizado	RUES y Google Maps API
Tiene ARAs (=1)	El ZAT tiene ARA(s) el periodo anterior al año analizado	RUES y Google Maps API
Controles variables por año		
Número de personas en el hogar por ZAT	Número promedio de personas en el hogar por ZAT en el año analizado	Encuestas de movilidad (2011, 2015, 2019, 2023)
Ingresos del hogar por ZAT	Ingreso promedio del hogar por ZAT en el año analizado	Encuestas de movilidad (2011, 2015, 2019, 2023)
Controles de línea base		
Habitantes por localidad en 2007	Número de habitantes por localidad en el 2007	Encuesta de Calidad de Vida de Bogotá 2007
Cantidad de personas por hogar 2007	Número promedio de personas en el hogar por localidad en el 2007	Encuesta de Calidad de Vida de Bogotá 2007
Indice de condiciones de vida 2007	Indice de condiciones de vida promedio por localidad en el 2007	Encuesta de Calidad de Vida de Bogotá 2007
Gasto promedio mensual 2007	Gasto promedio mensual en el hogar por localidad en el 2007	Encuesta de Calidad de Vida de Bogotá 2007
Controles fijos en el tiempo		

Table A.1: Data sources

Variable	Description	Data source
Estrato promedio del ZAT	Estrato promedio del ZAT, que oscila de 1 a 6	Mapas Bogotá - Infraestructura de Datos Espaciales de Bogotá (IDECA)
Estaciones de transmilenio cercanas	Cantidad de estaciones de transmilenio dentro o cercanos al ZAT	Mapas Bogotá - Infraestructura de Datos Espaciales de Bogotá (IDECA)
Cantidad acceso a vías arteriales	Cantidad de vías arteriales dentro o cercanos al ZAT	Mapas Bogotá - Infraestructura de Datos Espaciales de Bogotá (IDECA)
Control spillover		
Cantidad de Oxxos cercanos fuera del ZAT	Cantidad de OXXOs fuera del ZAT, que se encuentran cerca de este.	RUES y Google Maps API

Nota: No existe controles de línea base por ZAT, por lo que solo se pudo obtener índices previos al 2011 por localidad (ECDV 2007). El control "spillover" tiene en cuenta que hay ZATs que no tienen OXXOs, pero presentan OXXOs cercanos que pueden afectar la dinámica económica y laboral de la zona analizada. Finalmente, cuando de habla de OXXOs, vías arteriales y estaciones de transmilenio cercanos, significa que se encuentran a menos de 400m del ZAT.

B Evolution of OXXO stores in Bogotá divided by Transport Analysis Zones (ZATs)



(a) 2009 (b) 2011 (c) 2015

(d) 2019 (e) 2023 (f) 2025

Figure 5: **Evolution of the presence of OXXO by ZAT**

C Staggered controls of discount store chains

Table C.1: Means differences of hard discount store staggered controls for the 2018-4 cohort

	Still without treatment	**Treated**	**Means Difference**
Has D1s	0.477 (0.054)	0.477 (0.079)	0.172***
Quantity of D1s	1.233 (0.214)	3.250 (0.543)	2.017***
Has Aras	0.256 (0.047)	0.714 (0.087)	0.458***
Quantity of Aras	0.384 (0.089)	1.429 (0.264)	0.216***
Has J&B	0.895 (0.050)	1.964 (0.096)	1.069
Quantity of J&B	0.062 (0.256)	0.278 (1.202)	0.216***
UPZs	86	28	114

Nota: Significancia: *** p<0.01, ** p<0.05, * p<0.1.

Table C.2: Means differences of hard discount store staggered controls for the 2022-4 cohort

	Still without treatment	**Treated**	**Means Difference**
Has D1s	0.491 (0.067)	0.881 (0.042)	0.390***
Quantity of D1s	1.053 (0.191)	4.458 (0.512)	3.405***
Has Aras	0.298 (0.061)	0.627 (0.063)	0.329***
Quantity of Aras	0.386 (0.089)	1.712 (0.287)	1.326***
UPZs	57	59	116

Nota: Significancia: *** p<0.01, ** p<0.05, * p<0.1.